

RESEARCH ARTICLE

Purification of a recombinant oncolytic virus from clarified cell culture media by anion exchange monolith chromatography

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Funding information

AstraZeneca

Abstract

The use of viral vectors for vaccine, gene therapy, and oncolytic virotherapy applications has received increased attention in recent years. Large-scale purification of viral vector-based biotherapeutics still presents a significant technical challenge. Chromatography is the primary tool for the purification of biomolecules in the biotechnology industry; however, the majority of chromatography resins currently available have been designed for the purification of proteins. In contrast, convective interaction media monoliths are chromatographic supports that have been designed and successfully utilized for the purification of large biomolecules, including viruses, viruslike particles, and plasmids. We present a case study on the development of a purification method for recombinant Newcastle disease virus directly from clarified cell culture media using strong anion exchange monolith technology (CIMmultus QA, BIA Separations). Resin screening studies showed at least 10 times higher dynamic binding capacity of CIMmultus QA compared to traditional anion exchange chromatography resins. Design of experiments was used to demonstrate a robust operating window for the purification of recombinant virus directly from clarified cell culture without any further pH or conductivity adjustment of the load material. The capture step was successfully scaled up from 1 mL CIMmultus QA columns to the 8 L column scale and achieved a greater than 30-fold reduction in process volume. Compared to the load material, total host cell proteins were reduced by more than 76%, and residual host cell DNA by more than 57% in the elution pool, respectively. Direct loading of clarified cell culture onto a high-capacity monolith station-

Abbreviations: DBC, dynamic binding capacity; DOE, design of experiments; FFU, fluorescent focus unit; hcDNA, host cell DNA; rHCP, residual host cell protein; rNDV, recombinant Newcastle disease virus.

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ary phase makes convective flow chromatography an attractive alternative to centrifugation or TFF-based virus purification procedures.

KEYWORDS

anion exchange chromatography, dynamic binding capacity, large-scale manufacturing, monolith columns, Newcastle disease virus

1 | INTRODUCTION

Newcastle disease virus (NDV) is an avian paramyxovirus that naturally infects all species of birds [1]. Attenuated strains of NDV are of clinical interest as vaccine delivery vectors and for the treatment of cancers. NDV has been identified in preclinical models and clinical studies as a promising oncolytic agent for the treatment of a wide variety of tumors including leukemia, lymphoma, melanoma, neuroblastoma, fibrosarcoma, colon, lung, prostate, breast, gastric, and head and neck carcinoma [2]. Oncolytic viruses present a promising option for the treatment of tumors because they selectively replicate in and kill cancer cells without harming normal tissues and release tumor antigens that can stimulate an immune response [3–8]. T-VEC (Amgen) was the first and is currently the only FDA-approved oncolytic viral immunotherapy and is used for the treatment of metastatic melanoma [9].

NDV is a single negative strand RNA virus and belongs to the class of enveloped viruses. The lipid envelope is derived from the host cell plasma membrane and contains two NDV-specific transmembrane glycoproteins: hemagglutinin–neuraminidase (HN) and fusion protein (F). HN and F are both homo-oligomers and appear as spikes on the outer surface of the viral envelope. NDV ranges in size from 100 to 500 nm with a predominantly spherical morphology, although filamentous particles have been reported in the literature [10, 11].

The large size and diverse shape of NDV present a significant purification challenge. Furthermore, enveloped viruses such as NDV are fragile and sensitive to shear forces, and purification steps such as tangential flow filtration or column chromatography can cause the shedding of the viral envelope and lead to a loss of infectivity. For preclinical studies and most early-phase clinical trials, recombinant NDV (rNDV) has been produced in the allantoic fluid of embryonated chicken eggs [12–14]. For large scale and late-stage clinical trials, the generation of rNDV in cell culture is preferred. The successful production of high titer rNDV in HeLa, Vero, and HEK293 suspension cell cultures has been reported [15, 16]. Historically, purification of NDV has mostly been performed with a combination of filtration and sucrose density gradient ultracentrifugation steps (SGUC) [12–16]. A challenge

for SGUC is its limited scalability which presents a bottleneck for large-scale purification of NDV for clinical trials. More scalable purification steps such as membrane or column chromatography will be required to move NDV and other oncolytic viruses from clinic to market [17]. However, the size of many oncolytic viruses, including NDV, exceeds the pore diameter of most commercially available, bead-based chromatography resins which, as a result, leads to reduced binding capacities and the requirement for large chromatography columns. Despite these challenges, purifications of large, enveloped viruses including influenza A and B virus, measles, and hepatitis B virus with column chromatography have been reported [18–20]. In contrast, chromatography membranes provide large surface areas and channel diameters which results in high dynamic binding capacities (DBC), especially for large molecules and particles. Several viruses, including NDV, have been purified with anion exchange, cation exchange, and hydrophobic interaction chromatography membranes [21–24].

More recently, monolith-based chromatography has gained increased attention for the purification of large biomolecules. Monolith stationary phases have a unique architecture that consists of highly interconnected flow-through channels whose diameter can exceed more than 1000 nm. Monolith chromatography columns have several advantages over particle-based stationary phases, including high flow rates, low-pressure drops, reduced shear, high DBCs, and improved recovery for large biomolecules [25]. Anion and cation exchange and hydrophobic interaction monolith chromatography have been successfully applied to the purification of several different classes of virus, including measles, mumps, vaccinia, influenza, adenovirus, adeno associated virus, and HIV-1 gag VLPs [26–32].

Here, we report on anion exchange chromatography resin screening studies and the development of a capture step for recombinant NDV from clarified cell culture media by monolith chromatography. Anion exchange chromatography was chosen based on prior reports that diverse paramyxoviruses including influenza, mumps, and measles virus were purified with anion exchange chromatography [23, 26]. This is the first reported purification of cell culture derived, intact rNDV by column

chromatography. We expect that monolith-based chromatography will play an important role in bringing rNDV-based oncolytic virotherapies from the clinic to market.

2 | MATERIALS AND METHODS

2.1 | Materials

rNDV-containing cell culture medium was provided by the Cell Culture and Fermentation Sciences group at AstraZeneca. rNDV-containing cell culture medium was harvested for downstream processing by filtration with a positively charged depth filter and a 0.45 μm single-layer filter. All filters were equilibrated in 20 mM Tris and 100 mM NaCl prior to filtration. Post filtration, a recovery chase was performed with 20 mM Tris and 100 mM NaCl. The clarified cell culture medium was treated overnight with Benzonase (EMD Millipore) to digest residual host cell DNA (hcDNA) prior to loading onto chromatography columns. All chemical reagents were purchased from Sigma (St. Louis, MO, USA) and/or J.T. Baker (Phillipsburg, NJ, USA) unless stated otherwise. Different AEX resins were investigated as capture resins in this study. All resins were tested in a 1 mL prepacked column volume format. Poros XQ (Life Technologies, Grand Island, NY, USA), Fractogel SO3 (Merck KGaA, Darmstadt, Germany), Q Sepharose FF (GE Healthcare, Piscataway, NJ, USA), GigaCap Q-650M (Tosoh Biosciences, King of Prussia, PA, USA), UNOsphere Q (Bio-Rad, Hercules, CA, USA), and CIMmultus QA (BIA Separations).

2.2 | Purification

2.2.1 | Bench scale purification

All purification experiments were performed on AKTA Pure chromatography workstations (Cytiva). Resin and CIMmultus QA monolith (BIA Separations) screening experiments were performed at a residence time of 30 s, unless otherwise noted. Resin screening, process development, and characterization studies were performed as single-replicate experiments. Columns were equilibrated with five column volumes (CV) of 20 mM Tris, 100 mM NaCl, pH 7.5 prior to loading. Clarified and benzonase-treated CM was loaded directly onto the different 1 mL columns without any further adjustments. After loading, the columns were washed with 5 CV of the equilibration buffer. Elution was carried out by step elution with 20 mM Tris, 1 M NaCl, pH 7.5, unless otherwise noted. Columns were stripped with 2 M NaCl and sanitized with 0.5 M

NaOH. Contact time was a minimum of 60 min for each step, respectively. Load, flow through, eluate, and strip samples were collected and stored at -80°C prior to analysis. Load, flow, and eluate samples were analyzed for rNDV titer, residual host cell protein (rHCP), and hcDNA as described. Samples were directly analyzed without any further purification. Recovery (yield) was calculated based on the total amount of rNDV recovered in the elution pools compared to the total amount of rNDV in the load material.

2.2.2 | Dynamic binding capacity studies

DBC studies were performed using 1 mL prepacked columns. DBC studies as a function of monolith channel diameter were performed with 0.1 mL monolith columns to reduce the overall material requirements. DBC studies were performed as single-replicate experiments. Flow rates for each column were adjusted to maintain a 30 s residence time for all DBC experiments. DBC was determined by first equilibrating each column with 20 mM Tris 100 mM NaCl pH 7.5, then loading clarified NDV cell culture onto the AEX columns. Flow-through fractions were collected and analyzed for rNDV titer. Percent breakthrough was calculated by dividing the flow-through fraction titer (C) by the initial load titer (C_0). rNDV DBC capacity was estimated at 10% breakthrough.

2.2.3 | Large-scale purification

Large-scale purification was performed on AKTA Ready chromatography systems (Cytiva) with 8 L CIMmultus QA columns (BIA Separations) using the same process and buffers as described in Section 2.2.1. Scale up was performed by maintaining the volumetric column load challenges and residence times used at the 1 and 8 mL scale.

2.3 | Analytical methods

2.3.1 | rNDV infectious titer assay

rNDV titers were measured as described in Rush et al. [33]. Briefly, Vero cells were seeded in 96-well tissue culture plates to achieve 100% confluency on day of infection. Tenfold serial dilutions of samples were generated to achieve appropriate dilution for expected titer range. Four dilutions were added to the plates in duplicate to inoculate the cell monolayer. Plates were incubated for 16 h. Monolayers were fixed with 80% acetone and

immunostained using a polyclonal anti-NDV antibody (Abcam) and an Alexa Fluor 488 conjugated secondary antibody (Life Technologies). Plates were analyzed using an automated foci counting protocol on an ImageXpress Micro XLS Confocal (Molecular Devices) cell imager. rNDV infectious titer was reported in $\log_{10}$ fluorescent focus unit per mL ($\log_{10}$ FFU/mL). The variability of the assay was $\pm 0.2 \log_{10}$ FFU/mL.

2.3.2 | Residual host cell protein assay (rHCP)

rHCPs were measured with the HeLa HCP ELISA Kit (F810) from Cygnus Technologies (Southport, NC, USA). The HeLa Host Cell Protein assay is a two-site immunoenzymetric assay. Samples containing HeLa cell proteins are reacted in microtiter strips coated with an affinity-purified capture antibody. A second HRP labeled anti-HeLa antibody is reacted simultaneously, forming a sandwich complex of solid phase antibody-HeLa enzyme-labeled antibody. The microtiter strips are then washed to remove any unbound reactants. After the washes, the substrate tetramethylbenzidine is then reacted. The amount of hydrolyzed substrate is read on a SpectraMax M2 (Molecular Devices, San Jose, CA, USA) microtiter plate reader and is directly proportional to the concentration of HeLa Host Cell Proteins present. The lower limit of detection (LOD), defined as that concentration corresponding to a signal two standard deviations above the mean of the zero standard, is 0.5 ng/mL. The lower limit of quantitation, defined as the lowest concentration, where concentration coefficients of variation typically are <20%, is 2 ng/mL.

2.3.3 | Residual host cell DNA assay

A quantitative PCR assay is used to measure residual HeLa hcDNA in the test samples. This method uses forward and reverse primers targeting a unique and highly repetitive DNA sequence in the HeLa genome. Primer and probes were designed by an alignment of >1000 000 FASTA files returned by BLASTN of AH003462.2 (Human ALU repeat) against the human reference assembly to produce a conserved sequence suitable for primer design. Primers and probes were designed using primer3 and used at a concentration of 500, 500, and 250 nM (Fwd Primer, Rev Primer, and FAM labeled probe, respectively). Prior to analysis, samples are combined with chaotropic salts and then precipitated with isopropanol to isolate residual HeLa DNA. The HeLa DNA content is calculated based on a standard curve developed using HeLa genomic DNA. Results are reported as ng/mL. The lower LOD is 0.01 ng/mL.

2.4 | Design of experiment

Design of experiments (DOE) and statistical analysis of the results were performed with JMP 17 software (SAS, Carry, NC, USA). DOE experiments were performed with a single run per conditions.

2.4.1 | CryoTEM microscopy

CryoTEM microscopy was performed by NanoImaging Services (San Diego, CA, USA). Sample grids were prepared by preserving undiluted samples in vitrified ice supported by holey carbon films on 400-mesh copper grids. Each sample was prepared by applying a 3 μ L drop of sample suspension to a cleaned grid, blotting away with filter paper and immediately proceeding with vitrification in liquid ethane. Grids were stored under liquid nitrogen until transferred to the electron microscope for imaging. Electron microscopy was performed using an FEI Tecnai T12 electron microscope. Vitreous ice grids were transferred into the electron microscope using a cryostage that maintains the grids at a temperature below -17°C . Images of each grid were acquired at multiple scales to assess the overall distribution of the specimen.

3 | RESULTS AND DISCUSSION

Our goal was to develop a capture step for rNDV from clarified cell culture that did not require pH or conductivity adjustments. A key requirement, therefore, was robust purification performance over the expected pH and conductivity ranges of clarified cell culture. Multiple anion exchange resins were screened for their ability to bind rNDV under typical cell culture pH and conductivity conditions. DBC measurements were used to down select an anion exchange resin for further development and characterization, including elution conditions to recover rNDV. Process robustness was confirmed through DOE studies centered around the expected cell culture pH and conductivity conditions. Successful scale up and process performance were demonstrated during a 500 L scale clinical manufacturing campaign.

3.1 | rNDV capture step development from clarified cell cultures

rNDV was produced in suspension mammalian cell culture. Paramyxoviruses, such as NDV, assemble in and bud from plasma membranes of infected cells [34] and can be

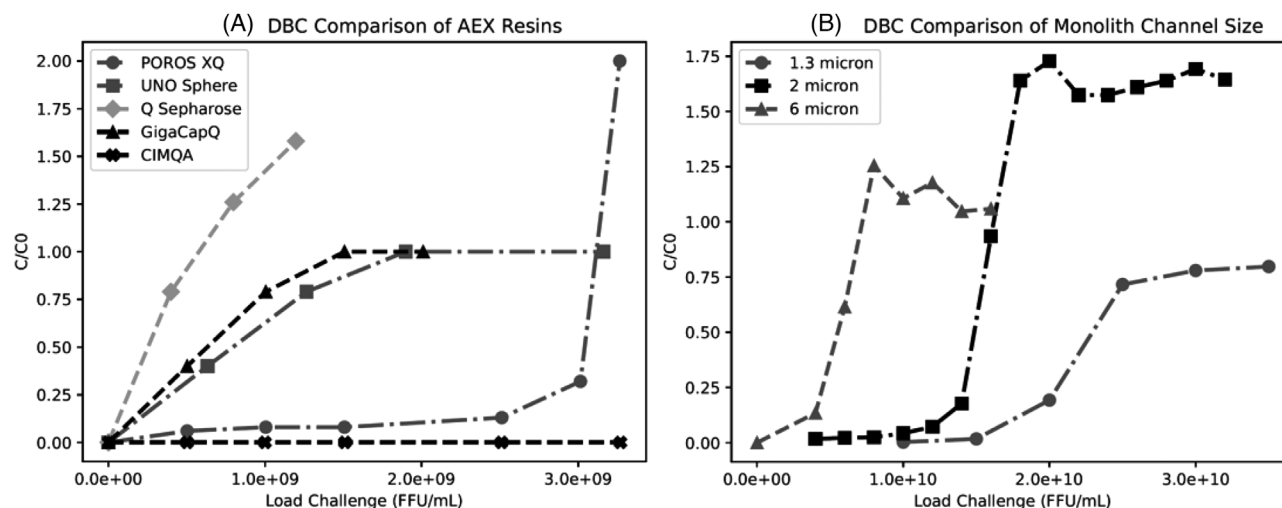


FIGURE 1 (A) Comparison of dynamic binding capacity (DBC) curves for clarified recombinant Newcastle disease virus (rNDV) cell culture loaded to 3.5E09 fluorescent focus unit (FFU)/mL on POROS XQ (●), UNOSphere (■), Q Sepharose (◆), GigaCapQ (▲) resins, and CIMmultus QA monolith (X). (B) Comparison of DBC curves for clarified rNDV cell culture loaded to 3.5E10 FFU/mL on 1.3, 2, and 6 μ m CIMmultus QA monolith. Data points shown are from single DBC chromatography experiments.

harvested from the cell culture supernatant by centrifugation or filtration, which removes intact cells and cell debris. Bioreactor conditions for rNDV have been optimized to maintain high cell viability at time of harvest and minimize the concentrations of rHCPs and hcDNA in the cell culture media. In addition, the endonuclease Benzonase was used to further reduce hcDNA after clarification, since it has been shown that residual hcDNA can interfere with the binding of viral particles to anion exchange resins [35].

3.2 | Anion exchange resin screening studies

Five AEX resins with differences in base matrices, pore diameters, and column designs were tested for their ability to bind rNDV. The goal was to evaluate the impact of these differences on process performance in terms of DBC. The following resins were screened: (1) Q Sepharose FF, (2) GigaCap Q-650M, (3) UNOSphere Q, (4) Poros XQ, and (5) CIMmultus QA (2 μ m). The first four of these resins were tested in a packed bed, axial flow format, whereas the CIMmultus QA resin was operated as a monolithic, radial flow chromatography column. Figure 1A shows that Q Sepharose FF, GigaCap Q-650M, and UNOSphere Q did not bind or only had a very low DBC for rNDV from clarified cell culture. rNDV was immediately detected in the early flow-through fractions. For each of these resins the DBC at 10% breakthrough was well below 1E9 FFU/mL. In contrast, POROS XQ had a DBC at 10% breakthrough of approximately 2.5E9 FFU/mL, whereas no rNDV was detected in any of the CIMmultus QA (2 μ m) fractions

up to a load challenge of 3.5E9 FFU/mL. These differences in DBC highlight an important consideration for the design of purification processes for large biomolecules such as enveloped viruses. The binding capacity of most commercially available chromatography resins designed for the purification of biomolecules such as proteins and antibodies resides in the internal pore volume of the chromatography bead. The average pore diameter of the Q Sepharose FF, GigaCap Q-650M, and UNOSphere Q resins is in the range of 50–125 nm, which is considerably smaller than the average size of rNDV (~200 nm). rNDV is therefore sterically excluded from the internal pore volume and can only interact with the small percentage of functional groups that are located on or close to the surface of the chromatography bead, resulting in the low DBC seen for these three resins in Figure 1A. The average pore diameter for POROS resins is in the range of 100–360 nm. As a result of this more open pore structure, a more than 10-fold increase in DBC was observed with POROS XQ compared to Q Sepharose FF, GigaCap Q-650M, and UNOSphere Q. In contrast, the CIMmultus QA column is designed as a precast anion exchange monolith with highly interconnected convective channels that are distributed homogeneously throughout the entire chromatography bed. For monolith columns channel size is defined as the average channel diameter within the porous methacrylate media. Large channels make it possible to purify large biomolecules including exosomes, viruses, and plasmids at high flow rates with low-pressure drops [36]. For the screening study CIMmultus QA column with a 2 μ m channel size was used. The absence of detectable rNDV in any of the fractions up to a load challenge of 3.5E9 FFU/mL

resins indicated a very high DBC of monolith columns for rNDV which demonstrated the importance of selecting the correct chromatography format for the purification of viral particles.

3.3 | CIMmultus QA binding capacity studies

CIMmultus QA monolith columns are available in three different channel sizes: 1.3, 2, and 6 μm . However, as the channel size increases, the internal surface area decreases with a potentially concomitant decrease in DBC. Based on data provided by the manufacturer, the internal surface areas for the 1.3, 2, and 6 μm monolith columns are in the range of 3.0–3.4, 1.8–2.1, and $\sim 0.8 \text{ m}^2/\text{mL}$ resin volumes, respectively. In order to determine the impact of channel size on rNDV DBC, binding capacity studies were repeated with a ten times higher load challenge of up to $3.5\text{E}10 \text{ FFU/mL}$ (Figure 1B). In order to minimize the amount of material required, these studies were conducted with 0.1 mL monolith columns maintaining the same residence time as used for the resin screening experiments. DBCs at 10% breakthrough for the 1.3, 2, and 6 μm CIMmultus QA column were $1.7\text{E}10$, $1.3\text{E}10$, and $0.3\text{E}10 \text{ FFU/mL}$, and demonstrated a nearly linear correlation between rNDV DBC and available surface area. Figure 1B also shows that the DBCs of the 1.3 and 2 μm CIMmultus QA monolith columns were approximately five- to sevenfold higher than the POROS XQ resin and more than 65-fold higher compared to the UNOsphere Q column. The high DBCs of the CIMmultus QA 1.3 and 2 μm columns for viral particles make it possible to capture and purify rNDV from large cell culture volume. Based on current cell culture titers, it was estimated that rNDV could be captured and purified from 500 to 1000 L of clarified cell culture with relatively small CIMmultus QA columns in the single-digit L scale. With three available monolith channel sizes, the 2 μm CIMmultus QA column was selected for further development and scale up. The selection was based on a compromise among DBC, product recovery, and pressure drop from the DBC studies. Although the 1.3 μm CIMmultus QA column had an approximately 24% higher DBC compared to the 2 μm CIMmultus QA column, the recovery of rNDV was more than twofold higher for the 2 μm CIMmultus QA column than for the 1.3 μm CIMmultus QA column as shown in Table S1. A potential explanation for the higher rNDV recovery observed on the 2 μm CIMmultus QA is the reduced likelihood of convective entrapment of viral particles and aggregates on the 2 μm monolith compared to the 1.3 μm monolith [36]. For the same reason, the 2 μm CIMmultus QA monolith column was considered to be

more advantageous for cleaning, sanitization, and reuse. Even though the 6 μm CIMmultus column had the lowest observed pressure drop during the DBC studies, which was considered an attractive feature for large-scale manufacturing, it was not considered further due to its fourfold lower DBC compared to the 2 μm CIMmultus QA column. At large scale, this would have required multiple chromatography cycles due to the fixed bed monolith column volume and would have resulted in extended hold times of the clarified cell culture with a potential loss of activity.

Both during the initial anion exchange resin screening experiments (Figure 1A) and the subsequent, channel diameter-dependent DBC studies (Figure 1B) higher than expected rNDV titer were observed in the early flow-through fractions of some of the tested resins, resulting in C/C_0 ratios of greater than one. A possible explanation may be that under overloaded conditions, rNDV particles either weakly bound to the stationary phase or retained as viral aggregates were displaced from the column leading to apparently higher titers than those measured in the initial starting material.

3.4 | CIMmultus QA development

In order to recover bound rNDV, the CIMQA monolith was first washed with 20 mM Tris 100 mM NaCl to displace bound rHCPs and hcDNA then eluted with a high salt 20 mM Tris 1 M NaCl buffer. During screening and selection of an appropriate elution buffer, it was observed the CIMQA performance was not sensitive to the elution buffer salt concentration. Tris of 20 mM with 750, 1000, 1250, or 1500 mM NaCl were evaluated, and minimal differences in rNDV recovery and rHCP reduction were observed (Figures S1 and S2). We selected elution with 1 M NaCl as this provided marginally higher recovery than our lower salt condition without increasing our rHCP concentration.

3.5 | Elution studies

A representative chromatogram demonstrating direct capture of rNDV from clarified cell culture media and elution with 20 mM Tris 1 M NaCl pH 7.5 on a 2 μm CIMmultus QA 1 mL monolith column is shown in Figure 2. Under the above-stated conditions, 37%–51% of rNDV was recovered in the product pool with a >95% reduction in rHCPs and >65% reduction in hcDNA (Table 1). Recovery of rNDV was within expectations since low to medium recovery of very large particles and enveloped viruses from anion exchange columns in the range of 15%–50%

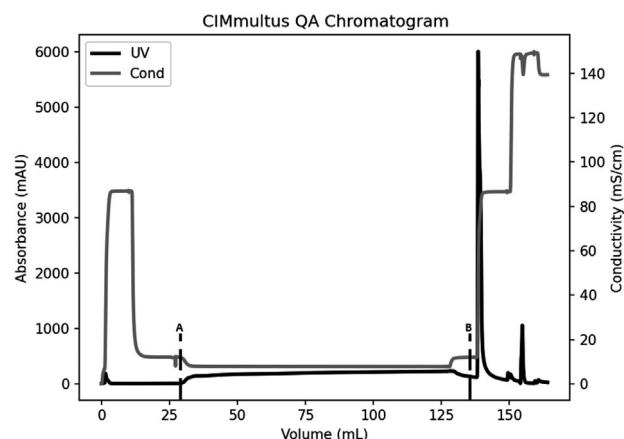


FIGURE 2 Chromatogram showing UV A280 and conductivity trace for clarified recombinant Newcastle disease virus (rNDV) cell culture loaded on 1 mL CIMmultus QA at 1E10 fluorescent focus unit (FFU)/mL. Label “A” identifies the beginning of load, label “B” identifies the beginning of elution.

TABLE 1 Recombinant Newcastle disease virus (rNDV) recovery, host cell protein clearance, and host cell DNA (hcDNA) clearance for three cycles of CIMmultus QA initial capture step at 1 mL scale.

Purification of rNDV with CIMmultus QA monolith over three process cycles			
Cycle	Clearance		rNDV recovery (%)
	hcDNA (%)	rHCP (%)	
1	64	96	51
2	73	96	48
3	67	96	37
Avg.	68	96	45

Abbreviation: rHCP, residual host cell protein.

has been reported previously [26, 32]. Less than quantitative recovery of rNDV may have been caused by on column aggregation of viral particles and entrapment on the column as the observed increase in back pressure during the loading phase suggests. Further, viral inactivation may have been caused by shear stress and surface interactions during the chromatography operation itself. More recently, the purification of NDV from the allantoic fluid of embryonated chicken eggs with anion exchange membrane chromatography in the presence of interference agents has been reported by Santry et al. NDV step yields in the range of 20%–25% were observed in the presence of 40–100 mM EDTA in the load material, whereas adding the same amount of citrate to the load material increased NDV step yields to greater than 70% with good rHCP and hcDNA clearance [21]. A direct comparison of the recovery results is difficult due the different nature of starting materials, purification devices, and titer assay used; however, the study by Santry et al. suggests that membrane chro-

matography may be a promising addition to the oncolytic virus purification toolbox if sufficient binding capacities can be established.

3.6 | Load condition robustness studies

Enveloped viruses such as oncolytic measles virus have been shown to be sensitive to mechanical stress conditions including those induced by agitation and pumping action through tubing, filters, and membranes [37, 38]. The elimination of a load conditioning step by titration, dilution, or tangential flow filtration would not only avoid exposure of rNDV to additional shear stresses and hold times but also simplify and accelerate large-scale manufacturing of clinical trial material. We therefore wanted to demonstrate that purification of rNDV directly from clarified cell culture with CIMmultus QA is possible without a load conditioning step and deliver robust performance in terms of recovery and clearance of process-related impurities within the expected load pH and conductivity ranges of the clarified cell culture material. pH and conductivity of clarified cell culture were found to be in the range of pH 7.4–7.6 and 9–13 mS/cm. In order to characterize the robustness of CIMmultus QA as an initial capture step, single and multifactorial studies were executed. To evaluate the impact of pH and conductivity on process performance, a two-factor central composite DOE study was designed and performed in the pH range 7.0–8.0 and conductivity range 5.0–17.0 mS/cm. Figure 3 demonstrates that robust impurity clearance with greater than 53% clearance of residual hcDNA and greater than 89% clearance of rHCP was achieved over a broad range of pH and conductivity conditions. rNDV recoveries were from 25% to 33% between pH 7.0 and 7.6 at conductivities greater than 8 mS/cm. The recoveries observed in this DOE study were similar to those observed in our process development studies (Tables S1 and S2).

The impact of flow rate on rNDV was evaluated at the 1 mL CIMmultus QA column scale as a single-factor study. A 10-fold increase in flow rate from 0.5 to 5 mL/min, corresponding to a residence time of 53 and 5 s, respectively, had no impact on the CIMmultus QA chromatography profile and rNDV recovery with observed yields being in the range of 40%–50% (Figure S2 and Table S3).

We observed in the course of our process development studies variable anion exchange chromatography step yields on the monolith columns with different lots of cell culture starting materials. One possible explanation is the complex composition of the load material. The study material not only contained process-related impurities, such as cell culture media components, rHCP, and hcDNA, but also of varying amounts of noninfectious particles and

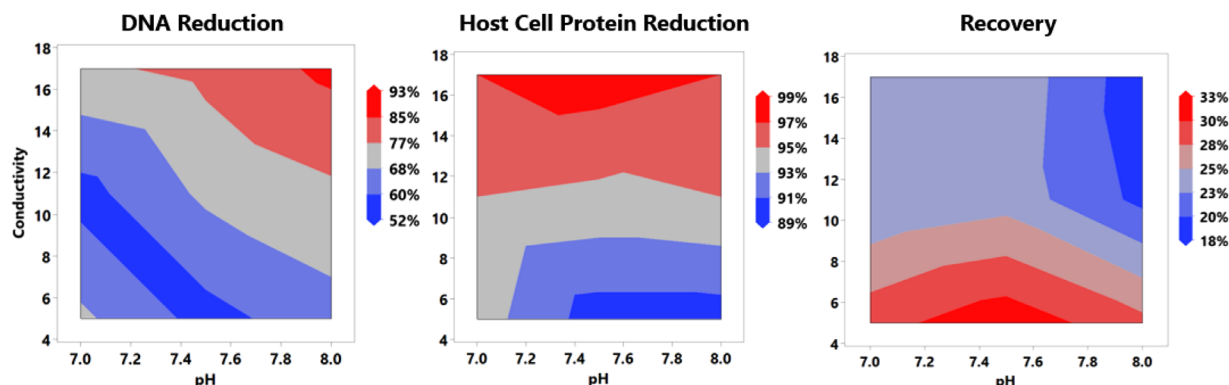


FIGURE 3 JMP 17 process robustness contour plots. Host cell DNA (hcDNA) and residual host cell protein (rHCP) clearance of CIMQA initial capture loaded at 1E10 fluorescent focus unit (FFU)/mL over range of load material pH (range 7.0–8.0) and conductivity (range 5–17 mS/cm).

exosomes that likely co-purified with intact rNDV particles and potentially impacted process performance [38]. A second challenge has been the incomplete mass balance of rNDV in our studies. Possible explanations for this observation are entrapment of viral particles in the channels of the monoliths, and viral inactivation through the interaction with the stationary phase during bind and elute conditions. Additional process development studies are needed to identify the root cause of incomplete mass balance and whether solution modifiers such as sugars, salts, and amino acids at non-inactivating concentrations can increase yields further.

Despite these challenges, our data demonstrates that rNDV can be effectively captured and purified from clarified cell culture without a load conditioning step and that the CIMmultus QA column is not sensitive to typical cell culture pH and conductivity variations within the investigated conditions (pH 7–8; conductivity: 5–17 mS/cm).

3.7 | Large-scale rNDV purification

Increased amounts of rNDV were required to support both preclinical and clinical studies. The cell culture process was scaled up from the single L scale to greater than 100 L scale. Likewise, the purification process was scaled up more than a 1000-fold from the 1 mL scale to the 8 L scale.

3.8 | Scale up consideration for monolith chromatography columns

Maintaining constant residence time is a well-established scale up strategy for column chromatography. This is typically achieved by keeping the column bed height and linear flow velocity constant between scales. The CIMmultus QA columns from BIA Separations are designed

as radial flow chromatography columns [39]. Large-scale monolith columns for preclinical and clinical production (0.08, 0.8, and 8.0 L) differ from small-scale process development columns (1 and 8 mL) in both their wall thickness and length. Due to the radial flow design, the linear flow rate inside the monolith columns is not constant and accelerates between inlet and outlet surfaces. For the purpose of scaling up the CIMmultus QA chromatography step from process development scale (1 mL) to clinical scale (8 L), residence time was calculated using the average linear velocity (u_{av}) across the wall according to Equation (1) where D_o and D_i are the outer and inner diameters of the monolith columns, respectively. The average linear velocity was calculated according to Equation (2) where F is the volumetric flow rate and L the monolith length.

$$\text{Residence time (RT)} = \frac{D_o - D_i}{2u_{av}} \quad (1)$$

$$\text{Average linear velocity (} u_{av} \text{)} = \frac{F}{\pi \times L} \frac{\ln\left(\frac{D_o}{D_i}\right)}{(D_o - D_i)} \quad (2)$$

Outer and inner diameters and monolith lengths for the different scales were provided by the BIA Separations. Equation (3) was used to calculate the volumetric flow rate with constant residence time at the 8 L scale column.

$$\text{Volumetric flow rate (} F \text{)} = \frac{L \times \pi}{4 \times RT} \frac{(D_o - D_i)^2}{\ln\left(\frac{D_o}{D_i}\right)} \quad (3)$$

3.8.1 | 8 L CIMmultus QA process performance

Direct loading of clarified cell culture and purification of rNDV with a 8 L CIMmultus QA monolith column was

TABLE 2 Recombinant Newcastle disease virus (rNDV) recovery, host cell protein clearance, and host cell DNA (hcDNA) clearance for five cycles of CIMQA initial capture step at 8 L scale.

Purification of rNDV with 8 L CIMmultus QA monolith			
Cycle	Clearance		rNDV recovery (%)
	hcDNA (%)	rHCP (%)	
1	50	82	54
2	57	76	27
3	60	83	54
4	76	84	25
5	75	86	26
Avg.	64 ± 10	82 ± 3	37 ± 14

Abbreviation: rHCP, residual host cell protein.

successfully implemented in a cGMP clinical manufacturing campaign at the 500 L bioreactor scale. The 8 L scale CIMmultus QA chromatography profiles were similar to those observed at the 1 mL CIMmultus QA scale chromatograms for five process cycles are shown in Figure S3. Table 2 shows that comparable process performance in terms of recovery and clearance of process-related impurities was achieved over five cycles with different cell culture lots. On average, a 64% reduction in residual hcDNA, and 82% reduction in rHCP was achieved. In addition, a more than 30-fold reduction in process volume was achieved in the initial capture step which facilitated rapid processing of the purified material to the drug substance stage. CryoTEM image in Figure S4 confirmed that our large-scale clinical manufacturing campaign produced rNDV within previously reported size and morphology [10, 11, 40–42].

4 | CONCLUDING REMARKS

Here, we report the first purification of rNDV by column chromatography. We show that the selection of the right stationary phase design is of key importance for the successful purification of enveloped viruses by column chromatography. The large channel diameter of the CIMmultus QA column enabled efficient binding of rNDV to the stationary phase whereas for most bead-based resin, rNDV was found in the flow-through fractions. Furthermore, we demonstrated that significant process simplification and acceleration could be achieved by direct loading of clarified cell culture material onto the anion exchange monolith column. Process performance of the CIMmultus QA column was robust over a wide range of cell culture pH and conductivity conditions, and a load conditioning step was not required. Successful scale up was demonstrated at the 8 L CIMmultus QA column scale and used to purify rNDV from over 500 L of clarified cell culture. It is expected that

monolith-based purification methods such as described here will enable large-scale manufacturing of rNDV to support late-stage clinical studies and bring rNDV to market as an oncolytic virotherapy agent.

ACKNOWLEDGMENTS

We would like to thank Ben Rush, George Chacko, and Jinlin Jiang for providing rNDV cell culture material. We would like to thank William Wang for his support and review of this manuscript. AstraZeneca provided funding for the conduct of this study.

CONFLICT OF INTEREST STATEMENT

The authors have declared no conflict of interest.

DATA AVAILABILITY STATEMENT

Data underlying the findings described in this article may be available from the corresponding author upon reasonable request in accordance with AstraZeneca's data sharing policy described at: <https://astrazenecagrouptrials.pharmacm.com/ST/Submission/Disclosure>.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

How to cite this article: Rogerson T, Xi G, Ampey A, Borman J, Jaroudi S, Pappas D, et al. Purification of a recombinant oncolytic virus from clarified cell culture media by anion exchange monolith chromatography. *Electrophoresis*. 2023;44:1923–1933.

<https://doi.org/10.1002/elps.202200270>